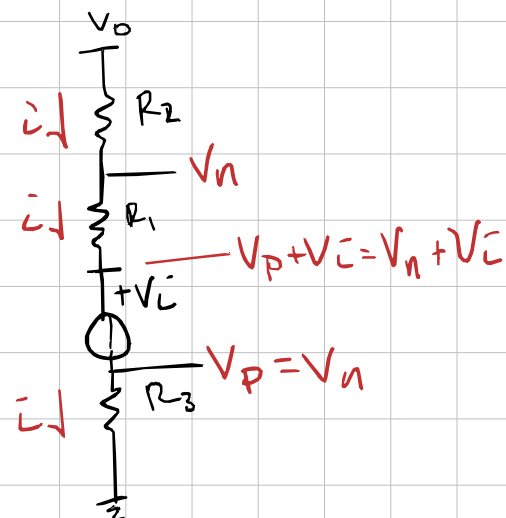
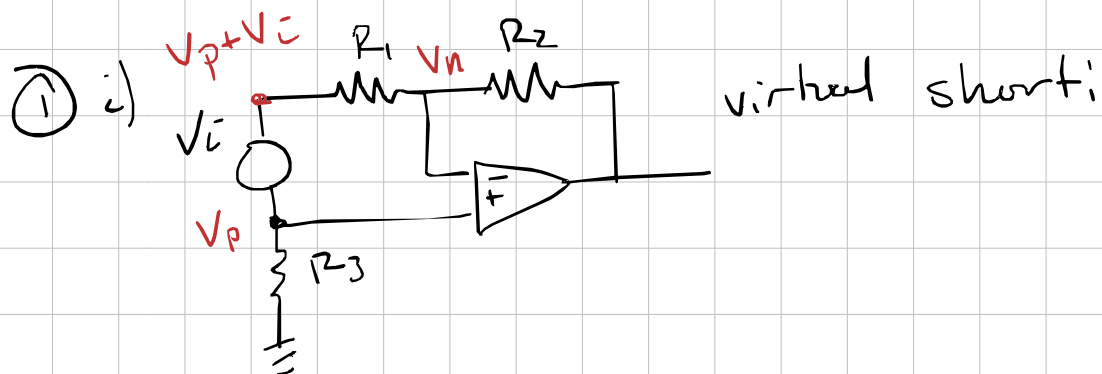




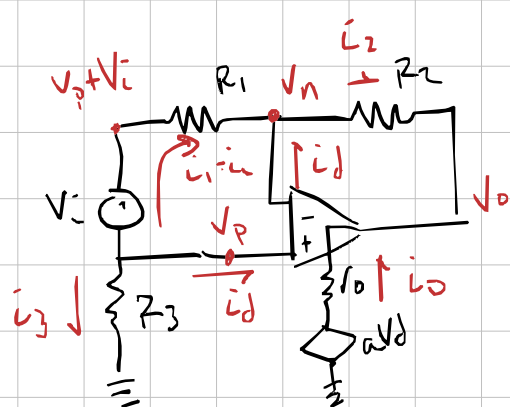
$$\begin{cases} V_p = V_n = i R_3 \\ (V_o - V_n) = i R_2 = V_o - i R_3 \\ V_n - (V_n + V_i) = i R_1 = -V_i \rightarrow i = -V_i / R_1 \end{cases}$$

$$-V_i \frac{R_2}{R_1} = V_o + \frac{V_i R_3}{R_1}$$

$$V_o = -\frac{V_i}{R_1} (R_2 + R_3) \rightarrow A_{ideal} = -\frac{R_2 + R_3}{R_1}$$

$$\therefore b = -\frac{R_1}{R_2 + R_3}$$

① $i_i \rightarrow$ Exact Analysis.



KCL @ V_p : $i_1 + i_3 + i_d = 0$ (1a)

$$\frac{V_p + V_i - V_n}{R_1} + \frac{V_p}{R_3} + \frac{V_p - V_n}{r_d} = 0$$
 (1b)

KCL @ V_n : $i_1 + i_d = i_2$ (2a)

$$\frac{V_p + V_i - V_n}{R_1} + \frac{V_p - V_n}{r_d} = \frac{V_n - V_o}{R_2}$$
 (2b)

Note from (1a) & (2a): $i_2 = -i_3$

$$\frac{V_n - V_o}{R_2} = -\frac{V_p}{R_3} \rightarrow V_p = \frac{R_3}{R_2} (V_o - V_n)$$
 (3)

KCL @ V_o : $i_o = -i_2$

$$\frac{aV_d - V_o}{r_o} = -\frac{V_n - V_o}{R_2} = \frac{V_o - V_n}{R_2}$$
 (4)

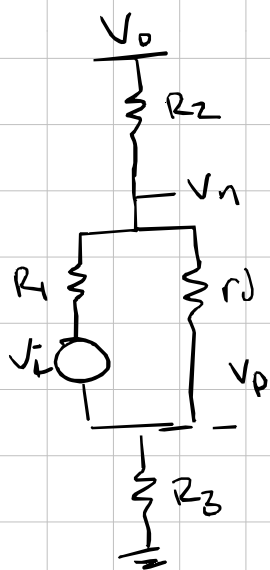
(4) $\rightarrow$ (3): $V_p = \frac{R_3}{r_o} (aV_d - V_o)$ (5)

(5) and $V_d = V_p - V_n \rightarrow$ (1b)

$$\frac{V_d + V_i}{R_1} + \frac{aV_d - V_o}{r_o} + \frac{V_d}{r_d} = 0$$

$$\frac{r_o}{R_1} V_i + V_d \left(\frac{r_o}{R_1} + \frac{r_o}{r_d} + a \right) = V_o$$
 (6)

Need to get rid of V_o ... find eq. for V_d with $V_i \hat{=}$ V_o

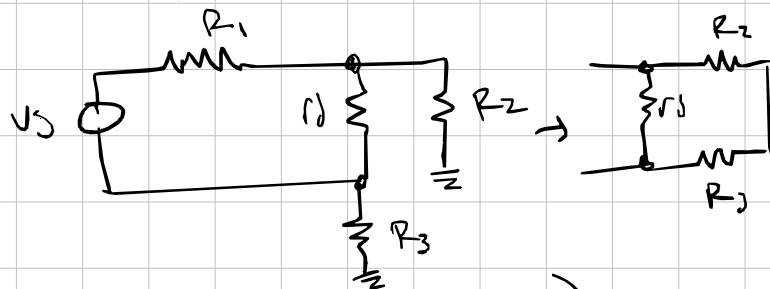


$\rightarrow$ Superposition:

$V_o \neq 0, V_i = 0$

$$V_d = -V_o \frac{R_1 || r_d}{R_1 || r_d + R_2 + R_3}$$

$V_o = 0, V_i \neq 0$



$$V_d = -V_i \left(\frac{r_d || (R_2 + R_3)}{r_d || (R_2 + R_3) + R_1} \right)$$

$$V_d = -V_o \frac{R_1 || r_d}{R_1 || r_d + (R_2 + R_3)(R_1 + r_d)} - V_i \left(\frac{r_d (R_2 + R_3)}{r_d (R_2 + R_3) + R_1 (r_d + R_2 + R_3)} \right)$$

dotted segments are equal!

$$V_d = - \left(R_1 V_o + (R_2 + R_3) V_i \right) / \left(R_1 + \left(1 + \frac{R_1}{r_d} \right) (R_2 + R_3) \right)$$
 (7)

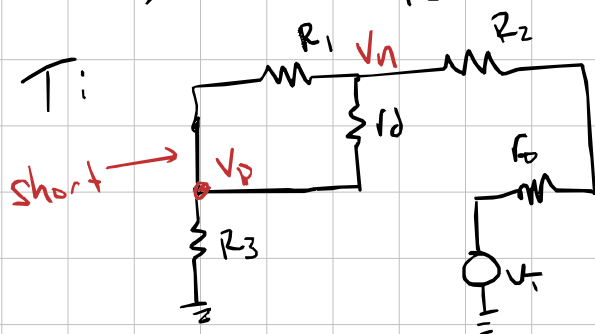
(7) $\rightarrow$ (6): $V_i \left[\frac{r_o}{R_1} - \frac{(R_2 + R_3) \left(\frac{r_o}{R_1} + \frac{r_o}{r_d} + a \right)}{R_1 + \left(1 + \frac{R_1}{r_d} \right) (R_2 + R_3)} \right]$

$$= V_o \left[1 + \frac{R_1 \left(\frac{r_o}{R_1} + \frac{r_o}{r_d} + a \right)}{R_1 + \left(1 + \frac{R_1}{r_d} \right) (R_2 + R_3)} \right]$$

$$\frac{V_o}{V_i} = \frac{\frac{r_o}{R_1} - \frac{(R_2 + R_3) \left(\frac{r_o}{R_1} + \frac{r_o}{r_d} + a \right)}{R_1 + \left(1 + \frac{R_1}{r_d} \right) (R_2 + R_3)}}{1 + \frac{R_1 \left(\frac{r_o}{R_1} + \frac{r_o}{r_d} + a \right)}{R_1 + \left(1 + \frac{R_1}{r_d} \right) (R_2 + R_3)}}$$

probably would simplify a bunch...

① $\bar{c}\bar{c}\bar{c} \rightarrow T \text{ i afe}$



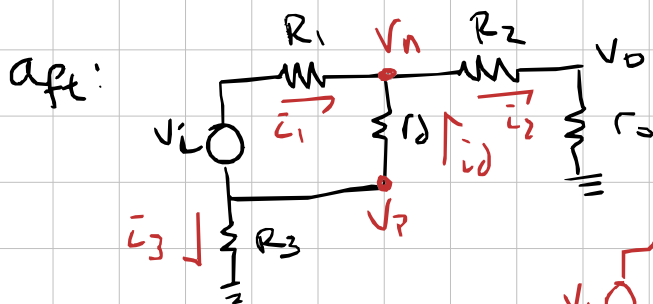
Just a large voltage divider...

$$V_d = V_p - V_n = -V_T \frac{R_1 || r_o}{r_o + R_2 + R_1 || r_o + R_3}$$

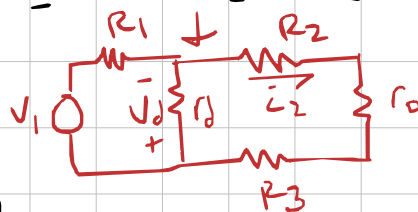
$$T = -\frac{a V_d}{V_T} = a R_1 || r_o / r_o + R_2 + R_1 || r_o + R_3$$

$$T = a R_1 r_o / R_1 r_o + (R_1 + r_o)(r_o + R_2 + R_3)$$

$$T = a / 1 + \left(\frac{1}{r_o} + \frac{1}{R_1} \right) (r_o + R_2 + R_3)$$



from previous,
 $i_2 = -i_3$



$$V_d = -V_i \frac{r_o || (R_2 + R_3 + r_o)}{r_o || (R_2 + R_3 + r_o) + R_1}$$

$$V_d = -V_i r_o (r_o + R_2 + R_3) / r_o (r_o + R_2 + R_3) + R_1 (r_o + r_o + R_2 + R_3)$$

$$\bar{i}_2 = -V_d / (r_o + R_2 + R_3)$$

$$V_o = \bar{i}_2 r_o = V_i r_o r_o / r_o (r_o + R_2 + R_3) + R_1 (r_o + r_o + R_2 + R_3)$$

$$\frac{V_o}{V_i} = \frac{r_o}{R_1} / 1 + \left(\frac{1}{r_o} + \frac{1}{R_1} \right) (r_o + R_2 + R_3) = a_{fe}$$

$$A = \frac{A_{fe} \bar{i}_2}{1 + 1/T} + \frac{a_{fe}}{1 + T}$$

① w)

$$r_o \rightarrow 0, R_D \rightarrow \infty$$

from Exact analysis:

$$\begin{aligned} \frac{V_o}{V_i} &= \frac{\frac{r_o}{R_1} - \frac{(R_2 + R_3) \left(\frac{r_o}{R_1} + \frac{r_o}{R_D} + a \right)}{R_1 + \left(1 + \frac{R_1}{R_D} \right) (R_2 + R_3)}}{1 + \frac{R_1 \left(\frac{r_o}{R_1} + \frac{r_o}{R_D} + a \right)}{R_1 + \left(1 + \frac{R_1}{R_D} \right) (R_2 + R_3)}} \\ &= \frac{-a(R_2 + R_3) / R_1 + R_2 + R_3}{1 + aR_1 / R_1 + R_2 + R_3} \\ &= - \frac{R_2 + R_3}{1 + \frac{R_1 + R_2 + R_3}{a}} \quad \checkmark \end{aligned}$$

from T_i afe

$$T = a / \left(1 + \left(\frac{1}{R_D} + \frac{1}{R_1} \right) (r_o + R_2 + R_3) \right)$$

$$T = a / \left(1 + \frac{1}{R_1} (R_2 + R_3) \right)$$

$$a_{fe} = \frac{r_o}{R_1} / \left(1 + \left(\frac{1}{R_D} + \frac{1}{R_1} \right) (r_o + R_2 + R_3) \right)$$

$$a_{fe} = 0$$

$$A_{ideal} = -R_2 + R_3 / R_1 \quad (\text{from i)})$$

$$A = \frac{A_{ideal}}{1 + 1/T} + \frac{a_{fe}}{1 + T}$$

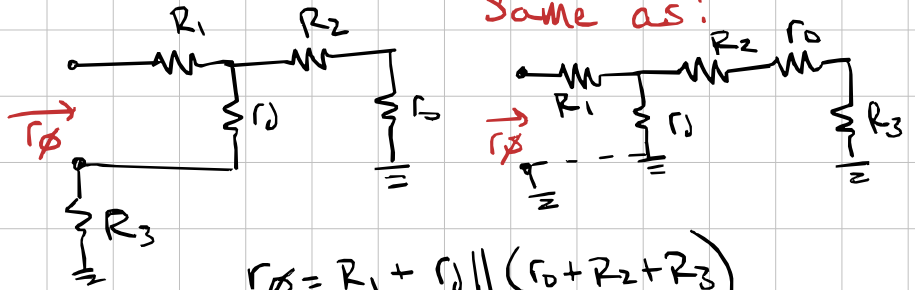
$$A = - \frac{R_2 + R_3 / R_1}{1 + \frac{1 + \frac{1}{R_1} (R_2 + R_3)}{a}}$$

$$A = - \frac{R_2 + R_3}{1 + \frac{R_1 + R_2 + R_3}{a}} \quad \checkmark$$

Note: $r_d \gg R_1, R_2, R_3 \gg r_o$ not applied until end. Does not necessarily need to be done this way!

① ✓

R_i r_ϕ :



$$r_\phi = R_1 + r_d \parallel (r_o + R_2 + R_3)$$

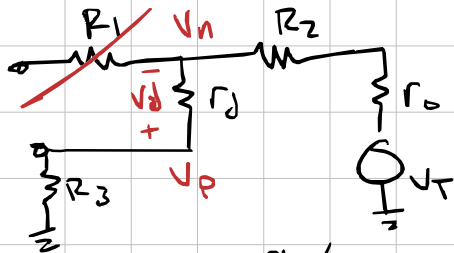
$$r_\phi = R_1 + r_d(r_o + R_2 + R_3) / (r_d + r_o + R_2 + R_3)$$

$$r_\phi = R_1 + \frac{r_o + R_2 + R_3}{1 + \frac{1}{r_d}(r_o + R_2 + R_3)}$$

T_{sc} : same as from $\bar{v}_i \bar{i}_i \rightarrow$

$$T_{sc} = a / 1 + \left(\frac{1}{r_d} + \frac{1}{R_1} \right) (r_o + R_2 + R_3)$$

T_{oc} :



$$v_d = -v_o \cdot r_d / (r_d + r_o + R_2 + R_3)$$

$$T_{oc} = a / 1 + \frac{1}{r_d} (r_o + R_2 + R_3)$$

$$R_i = \left[R_1 + \frac{r_o + R_2 + R_3}{1 + \frac{1}{r_d}(r_o + R_2 + R_3)} \right] \left[\frac{1 + a / 1 + \left(\frac{1}{r_d} + \frac{1}{R_1} \right) (r_o + R_2 + R_3)}{1 + a / 1 + \frac{1}{r_d} (r_o + R_2 + R_3)} \right]$$

Applying $r_d \gg R_1, R_2, R_3 \gg r_o$

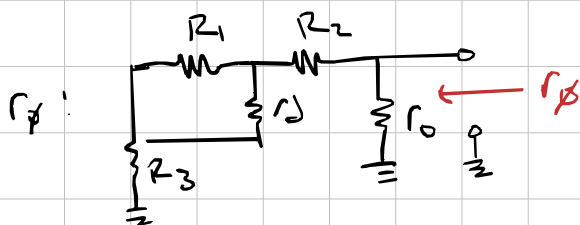
$$R_i = \left[R_1 + \frac{\cancel{r_o + R_2 + R_3}}{1 + \frac{1}{\cancel{r_d}}(\cancel{r_o + R_2 + R_3})} \right] \left[\frac{1 + a / 1 + \left(\frac{1}{\cancel{r_d}} + \frac{1}{R_1} \right) (\cancel{r_o + R_2 + R_3})}{1 + a / 1 + \frac{1}{\cancel{r_d}}(r_o + R_2 + R_3)} \right]$$

$\rightarrow 0$ $\rightarrow 0$

$$R_i \approx (R_1 + R_2 + R_3) \left[\frac{1 + a R_1 / (R_1 + R_2 + R_3)}{1 + a} \right]$$

① v)

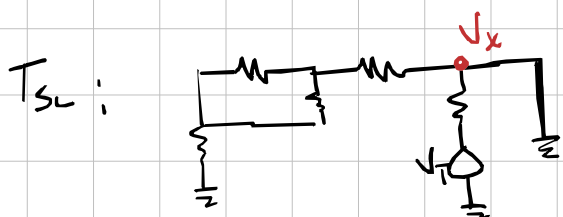
R_0



$$r_{\phi} = r_o \parallel (R_2 + R_1 \parallel r_o + R_3)$$

$$r_{\phi} = \frac{r_o ((R_2 + R_3)(R_1 + r_o) + R_1 r_o)}{(r_o + R_2 + R_3)(R_1 + r_o) + R_1 r_o}$$

$$r_{\phi} = \frac{r_o \left(1 + \left(\frac{1}{R_1} + \frac{1}{r_o} \right) (R_2 + R_3) \right)}{1 + \left(\frac{1}{R_1} + \frac{1}{r_o} \right) (r_o + R_2 + R_3)}$$



$$V_x \rightarrow 0, V_o \rightarrow 0$$

$$T_{SL} = 0$$

T_{OL} : same as from cii)

$$T_{OL} = a / \left(1 + \left(\frac{1}{r_o} + \frac{1}{R_1} \right) (r_o + R_2 + R_3) \right)$$

$$R_0 = \left[\frac{r_o \left(1 + \left(\frac{1}{R_1} + \frac{1}{r_o} \right) (R_2 + R_3) \right)}{1 + \left(\frac{1}{R_1} + \frac{1}{r_o} \right) (r_o + R_2 + R_3)} \right] \left[\frac{1}{1 + a / \left(1 + \left(\frac{1}{r_o} + \frac{1}{R_1} \right) (r_o + R_2 + R_3) \right)} \right]$$

Applying $r_o \gg R_1, R_2, R_3 \gg r_o$

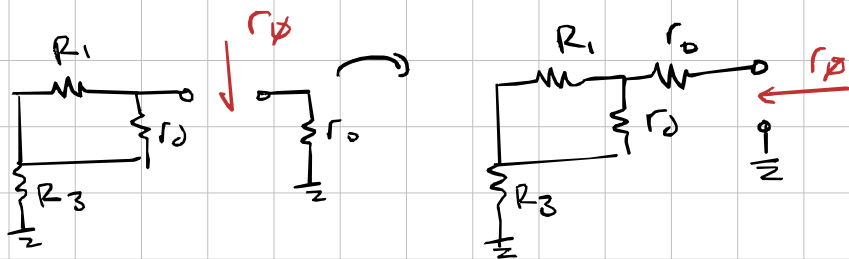
$$R_0 = \left[\frac{\cancel{r_o} \left(1 + \left(\frac{1}{R_1} + \frac{1}{\cancel{r_o}} \right) (R_2 + R_3) \right)}{1 + \left(\frac{1}{R_1} + \frac{1}{\cancel{r_o}} \right) (\cancel{r_o} + R_2 + R_3)} \right] \left[\frac{1}{1 + a / \left(1 + \left(\frac{1}{\cancel{r_o}} + \frac{1}{R_1} \right) (\cancel{r_o} + R_2 + R_3) \right)} \right]$$

$$R_0 = r_o / \left(1 + a R_1 / (R_1 + R_2 + R_3) \right)$$

if $r_o \rightarrow 0, R_0 \rightarrow 0$

①v)

R_{L2} r_o :

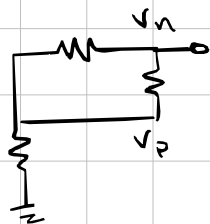


$$r_o = r_o + R_1 \parallel r_o + R_3 = r_o + R_3 + R_1 r_o / (R_1 + r_o)$$

T_{SL} : same as iii: Except $R_2 \rightarrow 0$

$$T_{SL} = a / 1 + \left(\frac{1}{r_o} + \frac{1}{R_1} \right) (r_o + R_3)$$

T_{OL} :



No connection to V_{IN} , $V_D \rightarrow$

$$V_D = 0, T_{OL} = 0$$

$$R_{L2} = \left(r_o + R_3 + R_1 r_o / (R_1 + r_o) \right) \left(1 + a / 1 + \left(\frac{1}{r_o} + \frac{1}{R_1} \right) (r_o + R_3) \right)$$

Applying $r_o \gg R_1, R_2, R_3 \gg r_o$

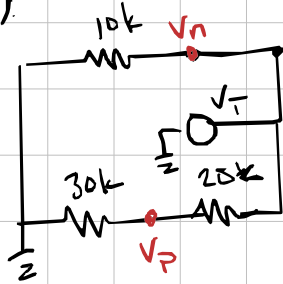
$$R_{L2} = \left(\cancel{r_o} + R_3 + \cancel{R_1 r_o / (R_1 + r_o)} \right) \left(1 + a / 1 + \left(\cancel{\frac{1}{r_o}} + \frac{1}{R_1} \right) (\cancel{r_o} + R_3) \right)$$

$\hookrightarrow R_1 r_o / r_o = R_1$

$$R_{L2} = (R_1 + R_3) \left(1 + a R_1 / (R_1 + R_3) \right)$$

$$R_{L2} = R_3 + R_1 (1 + a)$$

② Book 1.61: $a = 10^3 \text{ V/V}$, $r_d \rightarrow \infty$, $r_o \rightarrow 0$, find T .
P1.15)



$$V_d = V_p - V_n = V_p - V_T$$

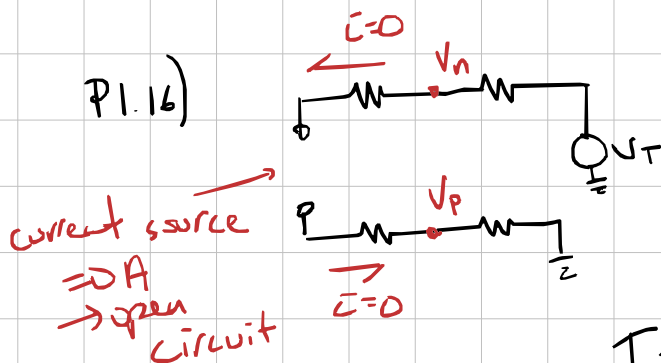
$$V_p = V_T \frac{30k}{20k+30k} = \frac{3}{5} V_T$$

$$V_d = \left(\frac{3}{5} - 1\right) V_T = -\frac{2}{5} V_T$$

$$T = -\frac{a V_d}{V_T}$$

$$= -10^3 \left(-\frac{2}{5} V_T\right) / V_T = \boxed{400}$$

P1.16)



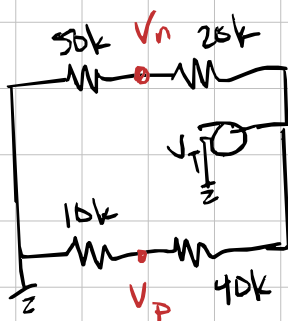
$$V_n = V_T$$

$$V_p = 0$$

$$V_d = -V_T$$

$$T = -10^3 (-V_T) / V_T = \boxed{10^3}$$

P1.17)



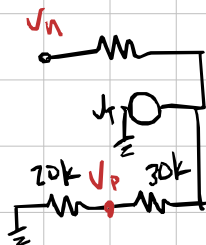
$$V_n = V_T \frac{50}{50+20} = \frac{5}{7} V_T$$

$$V_p = V_T \frac{10}{10+40} = \frac{1}{5} V_T$$

$$V_d = -0.514 V_T$$

$$T = 10^3 (0.514) = \boxed{514}$$

P1.18)



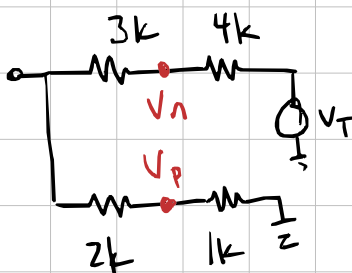
$$V_n = V_T$$

$$V_p = V_T \frac{20k}{20k+30k} = \frac{2}{5} V_T$$

$$V_d = -\frac{3}{5} V_T$$

$$T = 10^3 \left(\frac{3}{5}\right) = \boxed{600}$$

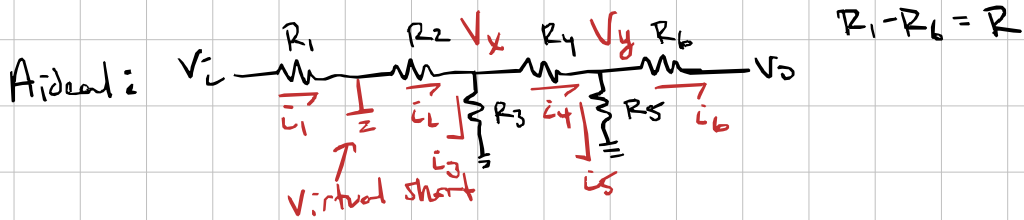
P1.19)



$$V_d = -V_T \frac{2k+3k}{4k+3k+2k+1k} = -\frac{1}{2} V_T$$

$$T = \boxed{500}$$

③ ELSE-60SD only



$$i_1 = i_2 \rightarrow \frac{V_i}{R} = \frac{0 - V_x}{R} \rightarrow V_x = -V_i$$

$$i_2 = i_3 + i_4 \rightarrow -\frac{V_x}{R} = \frac{V_x}{R} + \frac{V_x - V_y}{R}$$

$$3V_i = -V_g \rightarrow V_g = -3V_i$$

$$i_4 = i_5 + i_6 \rightarrow \frac{V_x - V_y}{R} = \frac{V_y}{R} + \frac{V_y - V_0}{R}$$

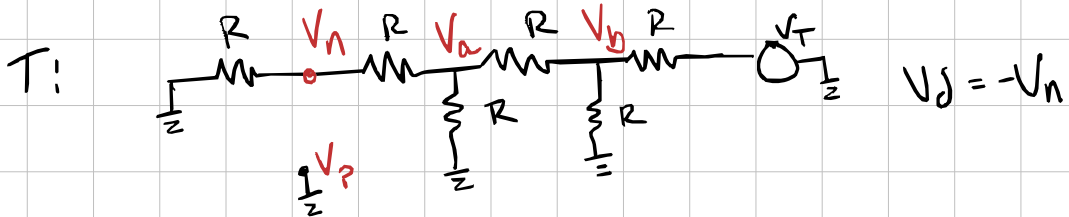
$$-V_i + 3V_i = -3V_i - 3V_i - V_o$$

$$A_{ideal} = -8$$

$$\leftarrow V_D = -8V_i$$

for Gain Error $\rightarrow$ find A

Since $r_D = 0$, $a_{gt} = 0$



$$V_b = R \parallel (R + R \parallel 2R) V_T / R \parallel (R + R \parallel 2R) + R$$

$$R + R \parallel 2R : R + \frac{2}{3}R = \frac{5}{3}R$$

$$R \parallel \frac{5}{3}R = \frac{5}{8}R$$

$$V_b = V_T^{5/8 / 13/8} = 5/13 V_T$$

$$V_a = V_b \frac{R \parallel 2R}{R + R \parallel 2R} = V_b \frac{2/3}{5/3} = \frac{2}{5} V_b = \frac{2}{13} V_T$$

$$V_n = V_a \cdot \frac{R}{2R} = \frac{1}{13} V_T \rightarrow T = \frac{a}{13}$$

$$A = \frac{A_{ideal}}{1 + 1/\tau} = -8 / (1 + 1/a)$$

$$\alpha = 10^3 \text{ V/V} \rightarrow A = -7.8973$$

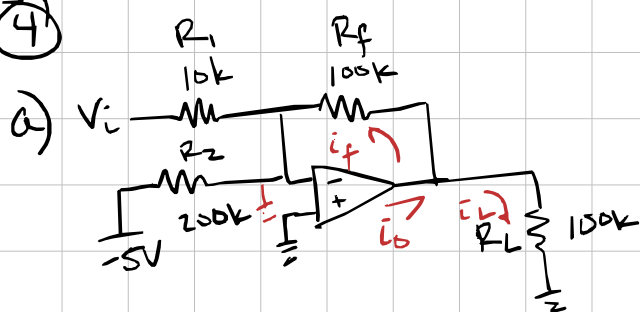
$$GE = \frac{-8 + 7.8973}{-8} \times 100\% = 1.28\%$$

b) $a_{\min}$ für $q_E \leq 0,1\% = 0,001$

$$0.501 = \frac{-8-A}{-8} \rightarrow A = -7.992 = \frac{-8}{1 + 13/a_{\min}}$$

$$a_{\min} = 12987\%$$

④



$$\bar{I}_g = 50 \mu A$$

$$\pm V_s = \pm 5V$$

Amp is inv. summer: $V_o = -R_f \left(\frac{V_i}{R_1} + \frac{-5V}{R_2} \right)$

$$V_o = -10V_i + 2.5V$$

$$\bar{I}_o = \bar{I}_f + \bar{I}_L = \frac{V_o}{R_f} + \frac{V_o}{R_L}$$

$$\bar{I}_o = \frac{2V_o}{100k}$$

$$P = P_g + P_o \quad P_g \text{ is fixed, so care about } P_o$$

Two regimes:

$$\bar{I}_o < 0 \quad (V_o < 0, V_i > 0.25V)$$

$$P_o = (-5V - V_o) \left(\frac{2V_o}{100k} \right) = -\frac{2V_o^2}{100k} - \frac{V_o}{10k}$$

find maximum:

$$\frac{\partial P_o}{\partial V_o} = -\frac{4V_o}{100k} - \frac{1}{10k} = 0 \rightarrow V_o = -2.5V$$

$$P_o = 125 \mu W$$

$$\bar{I}_o > 0 \quad (V_o > 0, V_i < 0.25V)$$

-- will be same since output is symmetric about ground

$$P_o = 125 \mu W \quad @ \quad V_o = 2.5V$$

$$P_g = (5V - -5V) \bar{I}_g$$

$$P_g = 500 \mu W$$

$$P_{max} = P_{omax} + P_g$$

$$P_{max} = 625 \mu W$$

$$\text{at } V_o = +2.5V \rightarrow V_i = 0V, \bar{I}_i = 0A$$

$$\bar{I}_o = 50 \mu A, \bar{I}_L = \bar{I}_f = 25 \mu A$$

$$\text{or } V_o = -2.5V \rightarrow V_i = 0.5V, \bar{I}_i = \frac{0.5}{10k} = 50 \mu A$$

$$\bar{I}_o = -50 \mu A, \bar{I}_L = \bar{I}_f = -25 \mu A$$

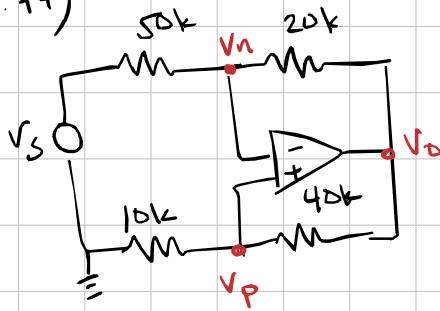
b) Need: $-12V \leq V_o \leq 12V$ what is V_i range?

$$-12V \leq -10V_i + 2.5V \leq 12V$$

$$-14.5V \leq -10V_i \leq 9.5V$$

$$1.45V \geq V_i \geq -0.95V$$

⑤ Book 1.77)



$$\pm V_{sat} = \pm 10V$$

$$V_n = (V_o - V_s) \frac{50}{20+50} + V_s$$

$$V_p = V_o \frac{10}{10+40}$$

for linear region, virtual short exists! $V_n = V_p$

$$\hookrightarrow -10V \leq V_o \leq 10V$$

$$(V_o - V_s) \frac{5}{7} + V_s = V_o \frac{1}{5} \rightarrow V_o \frac{5}{7} + V_s \frac{2}{7} = V_o \frac{1}{5}$$

$$A_{ideal} = \frac{V_o}{V_s} = -5/9$$

$$\rightarrow V_n = V_p = -\frac{1}{9} V_s$$

Non-linear (saturated region):

$$|V_s| > 10 / (5/9) = 18V$$

$$V_p = \pm 10 \frac{1}{5} = \pm 2V$$

$$V_n = \begin{cases} 7.14V + \frac{2}{7} V_s, & V_s < -18V \\ -7.14V + \frac{2}{7} V_s, & V_s > 18V \end{cases}$$

